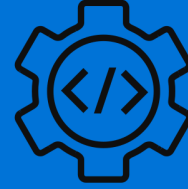


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MACHINE LEARNING CAPSTONE PROJECT - moyin.dev

Status **In progress** ▾

Timing Mar 15, 2026 to Mar 21, 2026

Owners Moyinoluwalogo Oladipo

OVERVIEW

This capstone introduces students to applied machine learning through realistic healthcare and pharmacy scenarios. Students will work with structured datasets designed to simulate operational data collected by digital health and pharmacy systems.

Participants must choose one of two tracks:

- 1 Linear Regression – Pharmacy Demand Forecasting
- 2 Logistic Regression – Medication Adherence Prediction

Each track represents a real-world problem currently faced by healthcare providers and digital health startups in Nigeria and other developing healthcare systems.

Students will be expected to apply the full machine learning workflow:

- Data understanding
- Exploratory Data Analysis
- Data preparation
- Model development

- Model evaluation
- Insight generation
- Real-world recommendations

The ultimate objective is not just to build a model, but to generate insights that could improve healthcare delivery and patient outcomes.

TRACK 1: LINEAR REGRESSION - PHARMACY DEMAND FORECASTING

Case Study

A mid-sized pharmacy network called HealthBridge Pharmacies operates multiple retail pharmacies across Lagos, including locations in Ikeja, Yaba, Surulere, and Lekki.

Over the past few years, the company has noticed that demand for antimalarial medications fluctuates significantly throughout the year. During some months, pharmacies experience shortages, while during others, they overstock medications that expire before being sold.

These inefficiencies have led to several operational problems:

- Stockouts during malaria outbreaks, forcing patients to visit competitors.
- Excess inventory, leading to drug expiration and financial losses.
- Poor demand forecasting, resulting in inefficient procurement decisions.

Management suspects that demand for antimalarial medications may be influenced by several factors, such as:

- Rainfall levels (mosquito breeding conditions)
- Seasonal malaria prevalence
- Drug pricing
- Competitor pricing
- Population density in the surrounding areas
- Advertising and promotional campaigns

To improve inventory planning, the pharmacy chain has collected historical operational data across multiple locations and months, capturing variables related to market conditions, environmental factors, and pricing strategies.

This dataset contains 1000 observations, representing monthly demand records across several pharmacy locations over multiple time periods.

The leadership team wants a predictive model that can estimate expected demand for antimalarial medications based on these influencing factors.

If accurate predictions can be generated, HealthBridge can:

- Procure drugs more efficiently
- Reduce expired inventory
- Ensure medications are available during high-demand periods
- Improve overall pharmacy profitability

Your task is to build a Linear Regression model that predicts monthly demand for antimalarial medications.

Objective

Develop a Linear Regression model that predicts:

Monthly_Demand_Units

Using the available explanatory variables.

Your model should identify the key drivers of medication demand and quantify how strongly each factor influences demand.

Key Questions

Your analysis should help the pharmacy answer the following strategic questions:

- Which variables most strongly influence medication demand?
- How do environmental factors such as rainfall and malaria prevalence affect drug sales?
- How sensitive is medication demand to pricing changes?
- Do competitor prices influence demand at HealthBridge pharmacies?

- Does population density influence medication demand?
- Can the pharmacy forecast demand accurately enough to improve stock planning?

Dataset Description

The dataset contains 1000 observations, each representing a monthly demand record for antimalarial medications across different pharmacy locations and environmental conditions.

Feature	Description
Advertising_Spend_Naira	Marketing expenditure used to promote medications
Drug_Price_Naira	Retail selling price of the medication
Competitor_Price_Naira	Average price charged by nearby pharmacies
Monthly_Rainfall_mm	Total rainfall in the area during the month
Malaria_Prevalence_Index	Estimated malaria prevalence in the region
Population_Density	Population density around the pharmacy location
Monthly_Demand_Units	Target variable representing units sold

Expected Impact

If implemented in real pharmacy operations, your model could help:

- Improve inventory planning
- Reduce stock shortages during malaria season
- Minimise expired medication losses
- Enable data-driven procurement decisions

TRACK 2: LOGISTIC REGRESSION - MEDICATION ADHERENCE PREDICTION

Case Study

Medication adherence is one of the largest challenges in healthcare systems worldwide.

Many patients fail to take their medications correctly due to factors such as:

- Forgetfulness
- Distance from pharmacies
- Cost of medications
- Lack of reminders
- Poor health education

Poor adherence can lead to serious consequences, including:

- Treatment failure
- Increased hospitalisations
- Development of drug resistance
- Higher healthcare costs

A digital health startup called MediTrack AI is developing a mobile platform designed to support patients in managing their medications.

The platform provides features such as:

- Medication reminders
- Pharmacy location services
- Patient adherence tracking
- Digital prescription management

To improve the platform, MediTrack AI wants to build a machine learning model that predicts which patients are most likely to fail to adhere to their medication schedules.

If such patients can be identified early, the system can automatically trigger interventions such as:

- SMS medication reminders
- Follow-up calls from pharmacists

- Adherence counselling
- Digital adherence monitoring

The company has collected 1000 anonymised patient records containing demographic information, behavioural indicators, and healthcare access variables.

Each record indicates whether the patient ultimately adhered to their prescribed medication regimen.

Your task is to build a Logistic Regression model that predicts the probability that a patient will adhere to their medication.

Objective

Develop a Logistic Regression model that predicts:

Adhered_To_Medication

Where:

- 1 = Patient adhered to medication
- 0 = Patient did not adhere

Your model should identify the key factors that influence medication adherence.

Key Questions

Your analysis should help MediTrack AI answer the following questions:

- Which patient characteristics are associated with medication adherence?
- Do SMS reminders significantly improve adherence?
- Does distance to the pharmacy reduce the likelihood of adherence?
- Does prior medication behaviour predict future adherence?
- Can we identify patients who may require additional support or intervention?

Dataset Description

The dataset contains 1000 patient records, each representing an individual patient's medication adherence outcome.

Feature	Description
Age	Patient age
Monthly_Income_Naira	Monthly income level
Distance_to_Pharmacy_km	Distance between patient home and pharmacy
SMS_Reminder	Whether the patient receives automated reminders
Previous_Adherence_Score	Historical medication adherence behavior
Chronic_Condition	Whether the patient has a long-term illness
Clinic_Visits_Last_Year	Number of clinic visits in the previous year
Adhered_To_Medication	Target variable (0 = No, 1 = Yes)

Expected Impact

If deployed in a real healthcare system, this model could help:

- Identify patients at risk of treatment failure
- Enable early intervention by pharmacists

- Improve patient adherence rates
- Reduce drug resistance and healthcare costs

Expected Deliverables

Students should submit a complete machine learning project containing:

1. Exploratory Data Analysis

- Summary statistics
- Data distributions
- Feature relationships
- Key patterns discovered

2. Data Preparation

- Data cleaning
- Feature engineering
- Handling missing values
- Train/test split

3. Model Development

- Linear Regression or Logistic Regression implementation
- Model training

4. Model Evaluation

For Linear Regression:

- R^2 Score
- RMSE

For Logistic Regression:

- Accuracy
- Precision
- Recall
- Confusion matrix

5. Insight Generation

Interpretation of:

- Feature importance
- Model coefficients
- Key drivers of predictions

6. Strategic Recommendations

Students must conclude with real-world recommendations based on their findings.

Example:

- Inventory planning improvements
- Patient adherence interventions

- Pharmacy pricing strategies
- Healthcare access improvements

FINAL NOTES

Submission: Submit all deliverables to the email moyin.dev@gmail.com .

Questions: Contact moyin.dev@gmail.com for clarifications.

Good luck! We're excited to see your submissions for this Machine Learning Capstone Project. Remember: the goal is not just to analyse data, but to provide actionable intelligence that drives real business value.

RESOURCES

Project brief  MACHINE LEARNING CAPSTONE PROJECT - moyin

Linear Regression Dataset  pharmacy_demand_regression_dataset_1000

Logistic Regression Dataset  medication_adherence_dataset_1000